

Dominik Pegler

PERSONAL

Born: October 18, 1982
Place: Vienna, Austria
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ORCID: [0009-0001-2345-1117](https://orcid.org/0009-0001-2345-1117)
Profession: Psychologist
Qualification: Master of Science (MSc)

EDUCATION

since 2023 **Doctoral Studies, Psychology**

University of Vienna

Dissertation Title: *Cognitive Alignment in Human–AI Interaction: Interpretability from Combinatorial Problems to Affective Vision*

Advisor: Prof. Frank Scharnowski

2021–2023 **Master of Science, Psychology**

University of Vienna

Thesis: *Predicting Psychological Embeddings of Fear-Related Stimuli in a Multidimensional Space Using Deep Neural Networks*

Advisor: Prof. Frank Scharnowski

(graduated with honors)

2015–2021 **Bachelor of Science, Psychology**

University of Vienna

EXPERIENCE

from July 2026 **TU Darmstadt, Germany**

Visiting researcher in the *Models of Higher Cognition* group (Prof. Frank Jäkel)

since August 2023 **University of Vienna, Austria**

University assistant (Pre-Doc) at the *Methods of Psychology* group (Prof. Frank Scharnowski)

February– June 2023	University of Vienna, Austria Project staff at the <i>Methods of Psychology</i> group (Prof. Frank Scharnowski) working on generalized assignment problems
2015–2023	Private sector, Austria Customer support and data analytics in manufacturing
April–May 2022	University of Vienna, Austria Research internship at the <i>Methods of Psychology</i> group (Prof. Frank Scharnowski) applying machine learning to psychological data
2003–2015	Private sector, Austria Production planning and controlling in manufacturing
August 2001	Private sector, Austria Summer internship technical support in telecommunications
July 2000	Private sector, Austria Summer internship research and development in manufacturing

TEACHING EXPERIENCE

2024-2025	University of Vienna, Austria Theory and Empirical Research 1 (Mind and Brain, Master Psychology) https://ufind.univie.ac.at/en/person.html?id=1001597 Responsibilities: Designed course curriculum and materials, guided students through various programming tutorials on statistics and machine learning, assessed student performance through grading assignments.
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PEER REVIEW

2026	IEEE Transactions on Affective Computing
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MEMBERSHIPS

since 2022	Austrian Psychological Society Österreichische Gesellschaft für Psychologie (ÖGP; www.oegps.at)
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GRANTS AND SCHOLARSHIPS

2026	Marietta Blau Grant Competitive funding by OeAD – Austria’s Agency for Education and Internationalisation – for a 6–12 month international research stay at TU Darmstadt (Prof. Frank Jäkel) (EUR 1,800/month).
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COMPUTER SKILLS

Python	PyTorch, pandas, NumPy, Matplotlib, scikit-learn, FastAPI/Flask
Javascript, HTML, CSS	React.js/Next.js, ReactNative
R	Tidyverse, lavaan
SQL	PostgreSQL, SQLite
Rust, C++, julia	Basics
Infrastructure	GNU/Linux, Docker, git
Other	L ^A T _E X, Emacs, JASP, Apache Superset, MS Access, Excel

LANGUAGES

German	Native
English	Fluent
Serbo-Croatian	Basic-Intermediate

INTERESTS

Cognitive Modeling, Reinforcement Learning, Human-Computer Interaction, Cultural Evolution, Interpretability, Computer Vision, Machine Learning, Clinical Research, Problem Solving, New Technologies, Interdisciplinary Projects, Learning, Cycling, Photography, Fiction

PUBLICATIONS (PEER-REVIEWED)

- Buchanan, E. M., Cuccolo, K., Heyman, T., van Berkel, N., Coles, N. A., Iyer, A., Peters, K., ... Pegler, D., ... Lewis, S. C. (2025). Measuring the semantic priming effect across many languages. *Nature Human Behaviour*, (2025), 1–20. <https://doi.org/10.1038/s41562-025-02254-x>

PUBLICATIONS (UNDER REVIEW)

- Pegler, D., Steyrl, D., Zhang, M., Karner, A., Arato, J., Scharnowski, F., & Melinscak, F. (2025). SpiderNets: Vision models predict human fear from aversive images. *Submitted to International Journal of Human-Computer Studies*. <https://doi.org/10.48550/arXiv.2509.04889>
- Pegler, D., Jäkel, F., Steyrl, D., Scharnowski, F., & Melinscak, F. (2026). Unpacking interpretability: Human-centered criteria for optimal combinatorial solutions. *Submitted to Computers in Human Behavior Reports*. <https://doi.org/10.48550/arXiv.2603.08856>
- Zhang, M., Pegler, D., Karner, A., Scharnowski, F., & Melinscak, F. (2025). Mapping mental representation of arachnophobia stimuli through similarity modeling. *Submitted to Journal of Affective Disorders*. https://doi.org/10.31219/osf.io/j8epm_v4

PUBLICATIONS (PREPRINTS)

- Karner, A., Obenaus, L., Zhang, M., Lor, C., Kistorz, K., Pegler, D., Leopold, M., Melinscak, F., Steyrl, D., Scharnowski, F. (2024). Visual attributes of spiders associated with aversiveness in spider-fearful individuals: A machine learning analysis. <https://doi.org/10.31234/osf.io/ht2pr>

CONFERENCE CONTRIBUTIONS

- Pegler, D., Steyrl, D., Scharnowski, F., & Melinscak, F. (2024, September). *Assessing human interpretability of machine-generated solutions to combinatorial design problems*. Congress of the German Psychological Society (DGPs), Vienna, Austria. [Contributed Talk] (Presented by D. Pegler)
- Zhang, M., Pegler, D., Scharnowski, F., & Melinscak, F. (2024, September). *Investigating mental representations of affective visual stimuli through the application of multidimensional scaling and convolutional neural networks*. Congress of the German Psychological Society (DGPs), Vienna, Austria. [Contributed Talk] (Presented by M. Zhang)
- Karner, A., Obenaus, L., Zhang, M., Lor, C., Kistorz, K., Pegler, D., Leopold, M.-L., Melinscak, F., Steyrl, D., & Scharnowski, F. (2024, September). *Visual attributes of spider images associated with aversiveness in spider-fearful individuals. A machine learning analysis*. Congress of the German Psychological Society (DGPs), Vienna, Austria. [Poster Presentation] (Presented by A. Karner)
- Zhang, M., Pegler, D., Scharnowski, F., & Melinscak, F. (2023, July). *Using multidimensional scaling and convolutional neural networks to probe mental representations of complex images*. MathPsych, Amsterdam, Netherlands. [Poster Presentation]. Retrieved from math-psych.org/presentation/1178. (Presented by M. Zhang)